J. Dairy Sci. 106:4918–4931
https://doi.org/10.3168/jds.2022-22876
© 2023, The Authors. Published by Elsevier Inc. and Fass Inc. on behalf of the American Dairy Science Association®.
This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Leukocytes, microRNA, and complement activity in raw, heat-treated, and frozen colostrum and their dynamics as colostrum transitions to mature milk in dairy cows

T. L. Chandler,[1] A. Newman,[1] J. E. Cha,[2] A. S. Sipka,[1] and S. Mann[1]*
[1]Department of Population Medicine and Diagnostic Sciences, College of Veterinary Medicine, Cornell University, Ithaca, NY 14853
[2]Division of Nutritional Sciences, College of Human Ecology, Cornell University, Ithaca, NY 14853

ABSTRACT

The objectives of this study were to evaluate the abundance and viability of leukocytes, the abundance of microRNA, and the activity of the complement pathway in (1) colostrum following heat-treatment or freezing, and (2) colostrum, transition milk, and mature milk. In experiment 1, composite colostrum samples were harvested from individual cows (n = 14) on a commercial dairy farm in NY and split into 3 aliquots using single-use colostrum bags. One aliquot was immediately cooled on ice following harvest (RAW) and stored at 4°C overnight, one was heat-treated for 60 min at 60°C (HT) before being cooled on ice and stored at 4°C overnight, and one was frozen at −20°C overnight (FR). The following morning, all samples were warmed to 40°C before further processing. In experiment 2, cows were sampled in a longitudinal study where composite samples were collected from colostrum (first milking, n = 23), transition milk (3 to 4 d postpartum, n = 13), and mature milk (6 to 7 d postpartum, n = 13). In both experiments colostrum was harvested from the first milking within 8 h of calving and samples were processed within 14 h of collection. Colostral leukocytes were isolated before viability was determined by trypan blue exclusion and manual differential cell counts were performed. Extracellular vesicles were isolated from whey by ultracentrifugation to isolate and quantify microRNA. Activity of the alternative complement pathway was determined in casein-depleted whey by semi-solid phase hemolysis assay. Somatic cell counts were determined for all raw samples. Macrophages and neutrophils made up the greatest proportion of leukocytes in colostrum followed by lymphocytes. Lymphocyte proportion increased as colostrum transitioned to mature milk, but overall somatic cell numbers declined concurrently. Viable cells were not isolated from HT or FR samples. Abundance of microRNA isolated from transition and mature milk was decreased compared with colostrum, did not differ between HT and RAW, but was increased in FR compared with RAW. Alternative complement pathway activity was decreased in HT, but not FR compared with RAW, and was not measurable in transition or mature milk. Postharvest heat-treatment and freezing of colostrum eliminated viable colostral leukocytes and affected microRNA abundance and complement activity. Leukocyte proportions, microRNA abundance, and complement activity changed as colostrum transitioned to mature milk. Although there were clear changes in the colostral components under study in relation to treatment and transition to mature milk, the biological significance of the described treatment effects and temporal changes were not investigated here.
Key words: colostrum, immune cell, complement, microRNA

INTRODUCTION

The importance of colostral Ig to bovine neonatal immunity is well established and research has mainly focused on the abundance of colostral IgG (Godden et al., 2019), but comparatively less is known about other bioactive factors that are also present in colostrum. Other immunologically active colostral components that have the potential to regulate immune function and development in calves have been identified in recent studies (Reschke et al., 2017; Mann et al., 2020; Van Hese et al., 2020). Colostral leukocytes (maternal white blood cells) could play active roles in both innate and adaptive immunity as shown in murine models (Cabinian et al., 2016; Ghosh et al., 2016). The role of bovine colostral leukocytes is not well characterized to date. MicroRNAs (**miRNA**) are also found in colostrum and are packaged in extracellular vesicles potentially making them available for activity within the neonate (Sun et al., 2013). We have recently characterized the low-abundant colostrum proteome and

Received October 7, 2022.
Accepted January 6, 2023.
*Corresponding author: sm682@cornell.edu

identified several proteins in the complement pathway (Mann et al., 2020) that are thought to play a crucial role in the neonate's innate immune defense (Korhonen et al., 2000, Ricklin et al., 2010). Of note, we identified several alternative complement pathway proteins (Mann et al., 2020) that are activated in the absence of antibodies and may have a specific benefit in the naïve bovine neonate

Interest in feeding colostrum or transition milk for an extended period after birth has grown because growth factors and other nutrients are still enriched compared with mature milk (reviewed by Godden et al., 2019). Immunologically active components may also be present in subsequent milkings but there is limited information on maternal leukocyte dynamics, miRNA abundance, and complement activity in colostrum and transition milk. To determine if these components are specifically enriched in colostrum compared with subsequent milkings, there is a need to further characterize changes in the first milkings postpartum.

Heat-treatment of colostrum is used on farm to reduce bacterial contamination and reduce the presence of certain infectious disease-causing organisms while maintaining the important IgG component (Mann et al., 2020; Donahue et al., 2012). Other nonprotein immunological factors in colostrum may be sensitive to heat-treatment but this has not been demonstrated. Freezing colostrum, with or without heat-treatment, is also used to help maintain colostrum supply, but is known to eliminate viable leukocytes (Donovan et al., 2007), has the potential to decrease complement pathway protein abundance (Mann et al., 2020), and may disrupt the integrity of microRNA.

Current recommendations for colostrum feeding, management, and storage are typically based on supplying the calf with adequate amounts of IgG, but have not taken into account the influence of postharvest storage and treatment on other immunologically active factors such as colostral leukocytes, miRNA, and complement activity with potential importance to the calf. Furthermore, the dynamics of these immunologically active factors as colostrum transitions to mature milk are not clear. We hypothesized that heat-treatment and freezing would alter the abundance of colostral leukocytes, miRNA, and complement activity compared with raw colostrum, and that these factors would be altered in abundance as colostrum transitions to mature milk. Therefore, the objectives of this study were to evaluate the abundance and viability of leukocytes, the abundance of microRNA, and the activity of the complement pathway in (1) colostrum following heat-treatment or freezing, and (2) colostrum, transition milk, and mature milk.

MATERIALS AND METHODS

Animals and Sampling

Experiment 1. All procedures were evaluated and approved by the Cornell University Institutional Animal Care and Use Committee (protocol no, 2018–0021; Ithaca, NY). The study was performed between September and October of 2021 on a commercial dairy farm in New York State. Farm-specific cow management and colostrum harvest procedures have been previously described (Mann et al., 2020). Briefly, farm personnel collected first-milking colostrum into graduated sanitized buckets 3 times per day. For this study, colostrum was harvested only at the evening milking (1900 h) from Holstein cows (n = 14) entering their first to sixth parity. Immediately upon collection, colostrum yield was measured and research personnel gently mixed colostrum with a whisk before taking an aliquot with a disposable transfer pipet to test Brix percentage on a digital refractometer (Palm Abbe, Misco) according to the manufacturer's instructions. Colostrum ≥20% Brix and ≥2 L of total volume was eligible to be used in the study. Colostrum (n = 14) from each individual cow was mixed again carefully by research personnel and then aliquoted into 3 separate 2-L disposable bags (Perfect Udder, Dairy Tech Inc.).

Bags were sealed before colostrum aliquots were subjected to 1 of 3 treatments: (1) bags were immediately placed on ice (**RAW**, n = 14) and stored overnight at 4°C, (2) bags were cooled on ice and frozen at −20°C overnight (**FR**, n = 14), or (3) bags were heat-treated for 60 min at 60°C (**HT**, n = 14) before cooling on ice and storage overnight at 4°C. A commercial pasteurizer (Dairy Tech Inc.) was used for batch heat-treatment of colostrum collected from animals on the same day and run on the "Colostrum Normal" profile as previously described (Mann et al., 2020). When the water bath reached the programmed cool-down temperature of 43°C (approximately 25 min after the completion of the heat-treatment cycle), the HT bags were removed and immediately placed on ice to rapidly cool before storage at 4°C overnight. The following morning, within approximately 14 h after harvest, all bags were warmed to 40°C in a water bath maintained at 40°C and colostrum in bags was thoroughly mixed before aliquots were taken for analysis and processing (described below). An aliquot of colostrum from each treatment was submitted to the Quality Milk Production Services Laboratory in the New York State Animal Health Diagnostic Center (Cornell University, Ithaca, NY) to identify and quantify bacteria by plate culture as previously described

(Mann et al., 2020). Colostrum Brix percentage was measured in RAW, HT, and FR after warming to 40°C on a digital refractometer (Palm Abbe, Misco).

Experiment 2. All procedures were evaluated and approved by the Cornell University Institutional Animal Care and Use Committee (protocol number 2020–0030). Postpartum Holstein cows entering their second or greater lactation that did not experience dystocia or had clinical signs of mastitis were enrolled in this longitudinal study from the herd at the Cornell University Ruminant Center (Harford, NY). Cows were enrolled and samples and data collected between August and October 2020.

During the sampling period, colostrum was collected from cows (n = 23) entering their second to fourth parity, and longitudinal samples that included colostrum, transition milk, and mature milk from the same animal were collected from a subset of cows (n = 13) entering their second or third parity. Selection criteria for this subset included availability of cows for daily milk sampling and monitoring. Colostrum was harvested according to the farm protocol at the first milking within 8 h after parturition. Briefly, colostrum was harvested from individual cows in a double-16 parallel parlor (DeLaval) and colostrum collected into a sanitized bucket. Colostrum was carefully mixed and sampled to measure Brix percentage on a digital refractometer (Palm Abbe, Misco) immediately after harvest. Within a few days after calving, animals were moved to individual sawdust-bedded tie stalls, continued on the same fresh cow diet, and milked twice daily using a portable-unit bucket milker (MMU22 Mobile Milking Unit, DeLaval). Transition milk was collected 3 to 4 d postpartum, and mature milk was collected 6 to 7 d postpartum during the morning milking at 0530 h. Composite colostrum and milk were weighted immediately after harvest and mixed carefully with a whisk before a 250-mL sample was collected and processed as described below. Colostrum samples were stored at 4°C for less than 14 h if not processed immediately, in which case samples were kept at ambient temperature for transport to the lab. Colostrum that was stored at 4°C was allowed to come to ambient temperature (20 to 22°C) for 30 min before processing. Transition and mature milk samples were kept at ambient temperature before processing and were processed within 4 h of harvest. A second aliquot of colostrum or milk was diluted 2-fold with phosphate buffered saline (PBS, pH 7.4) and stored at 4°C with bronopol preservative for analysis of somatic cell count using optical fluorescence on a Fossomatic FC (method 972.160; AOAC International, 2012) at Dairy One Cooperative Inc. within 1 wk of collection.

Sample Processing and Analysis

Whey Isolation. To de-fat samples, colostrum and milk samples were filtered through gauze before 40 mL of colostrum or milk was diluted 2-fold with PBS in 50-mL conical tubes and centrifuged at 4,000 × *g* for 30 min at 4°C. The fat layer was removed, and whey was mixed and aliquoted into either a 15-mL conical tube for later miRNA isolation and extraction, or a 4-mL cryovial for later complement pathway activity and stored at −80°C.

Leukocyte Isolation and Analysis. Leukocytes were isolated from colostrum and milk by centrifugation based on the method of Meganck et al. (2014) with modification. Cell isolation was carried out at ambient temperature. Samples were filtered through gauze before 20 mL of colostrum was diluted 5-fold and 200 mL of milk was diluted 2-fold using room-temperature PBS in 50-mL conical tubes. Samples were mixed by gentle inversion to avoid homogenization of milk fat and cell damage. Cells were pelleted at 550 × *g* for 15 min at 15°C. The fat layer was removed and the supernatant was discarded. The cell pellet was dispersed with 1 mL of PBS by gentle pipetting and the cell suspension was transferred to a clean 50 mL conical tube and cells were washed twice in 30 mL of PBS (300 × *g* for 5 min at 15°C). The supernatant was discarded and the cell pellet was dispersed with 1 mL of PBS before the cell suspension was filtered through a 40-μm nylon cell strainer (VWR International, LLC). Cell suspensions were pooled and washed with 30 mL of PBS (300 × *g* for 5 min at 15°C). If obvious red staining of the cell pellet was noted indicative of the presence of red blood cells, red blood cells were lysed using cold aluminum-chloride-potassium red cell lysis buffer (ACK buffer, Lonza) and washed twice with PBS (300 × *g* for 5 min at 15°C). Final cell pellets were resuspended in 5 mL of PBS before cell viability was determined by trypan blue exclusion. Cytocentrifuge preparations were made from 5×10^4 cells diluted in 100 μL of PBS (Shandon Cytospin 2, Marshall Scientific). Manual differential cell counts were performed on Wright's stained smears (Hema-Tek 1000) by a board-certified clinical pathologist (Animal Health Diagnostic Center, Ithaca, NY). A 200- cell differential was performed on slides when sample cellularity allowed, and results averaged. For lower cellularity samples, a 100-cell differential (or less if fewer than 100 cells present) was performed. Cytology images were obtained (Olympus UC90) from cytocentrifuge preparations.

MicroRNA Isolation and Analysis. Extracellular vesicles and microRNA were isolated based on the method of Izumi et al. (2015) with modification. Whey

samples were thawed for 30 min in water at 10°C before 12 mL of sample was pipetted into an ultracentrifuge tube (#344058, Ultra-Clear 25 × 89 mm, 38.5 mL) and 360 μL of cold 10 m*M* phenyl-methyl-sulfonyl fluoride, stored on ice, was added to inhibit proteolysis along with 6 mL of cold nuclease-free PBS. To avoid casein gel formation, 18 mL of cold 2% sodium citrate was added. Samples were centrifuged at 21,000 × *g* for 1 h at 4°C to remove residual fat, casein, and cell debris. Residual fat was aspirated off before the supernatant was carefully pipetted off and successively filtered through 0.45 and 0.22-μm syringe filters. Extracellular vesicles were sedimented by ultracentrifugation by diluting 26 mL of the filtered supernatant from above with 10 mL of cold nuclease-free PBS and centrifuged at 100,000 × *g* for 90 min at 4°C. The supernatant was removed, and the pellet resuspended in 300 μL of cold nuclease-free PBS and 90 μL of room-temperature buffer RPL (miRNeasy Serum/Plasma Advanced Kit, Qiagen). Samples were mixed by pipetting and transferred to a clean nuclease-free microcentrifuge tube and stored at −20°C until miRNA extraction.

MicroRNA was isolated using a miRNeasy Serum/Plasma Advanced Kit (Qiagen) according to the manufacturer's instructions. Total miRNA in samples was quantified by fluorescence using the Quant-iT microRNA Assay Kit (Life Technologies, Thermo Fisher Scientific). Samples were quantified in duplicate with a fluorescence microplate reader (BioTek Synergy 2, Agilent Technologies) according to the manufacturer's instructions. To further characterize isolated microRNA from longitudinal samples collected in experiment 2, an equal volume of each sample was pooled for colostrum, transition milk, and mature milk, and RNA was characterized using a Tape-Station system and RNA Screentape Assay (Agilent Technologies).

MicroRNA isolated from longitudinal samples collected in experiment 2 was further characterized by PCR. A volume containing 0.5 ng of miRNA was reverse transcribed to cDNA and pre-amplified using a TaqMan Advanced microRNA cDNA synthesis kit (Applied Biosystems, Thermo Fisher Scientific). A sample was reserve transcribed and pre-amplified in parallel to serve as the calibrator sample. A synthetic RNA oligo (cel-miR-39–3p: UCACCGGGUGUAAAUCAGCUUG) with a 5′ phosphate sense modification was included in the reaction at a concentration of 5.4 p*M* in the sample as an exogenous control. Primer probe sets (TaqMan Advanced microRNA assays, Applied Biosystems, Thermo Fisher Scientific) were purchased for microRNA targets of interest (miR-148a-3p: UCAGUGCACUACAGAACUUUGU, miR-27b-3p: UUCACAGUGGCUAAGUUCUGC, miR- let-7a-5p: UGAGGUAGUAGGUUGUAUAGUU, miR-26a-5p: UUCAAGUAAUCCAGGAUAGGCU) and the exogenous control. Real-time reverse-transcription quantitative PCR was performed in triplicate using a 1:10 dilution of cDNA at 25% of the final 10 μL reaction volume using TaqMan Fast Advanced Master Mix (Applied Biosystems, Thermo Fisher Scientific) in a QuantStudio 12K Flex Real-Time PCR System (Applied Biosystems, Thermo Fisher Scientific). The exogenous control was amplified in all samples at the same time as microRNA targets. A pool of microRNA from all samples was included as a no-reverse-transcription control and water served as a no-template control. Run conditions for all microRNA of interest were 1 cycle at 95°C for 20 s, followed by 40 cycles of 95°C for 1 s and 60°C for 20 s. Results of raw cycle thresholds (**CT**) for microRNA targets and the exogenous control were exported for statistical analysis.

Alternative Complement Pathway Activity. To deplete casein, whey samples were thawed for 30 min in cold water before 4 mL of whey was pipetted into an ultracentrifuge tube (#344058, Ultra-Clear 25 × 89 mm, 38.5 mL) and samples centrifuged at 100,000 × *g* for 45 min at 4°C. Residual fat was removed and the supernatant aliquoted into 1-mL cryovials for storage at −80°C until complement assay. The activity of the alternative complement pathway was measured using a semi-solid phase hemolysis assay described by Kerboua and Djenouhat (2020) with modifications. A 2% solution of agarose was made by melting 2 g of low-melting point agarose (UltraPure, Invitrogen) in 100 mL of deionized water. Guinea pig red blood cells (RBC, Innovative Research) were washed (500 × *g* for 10 min) in Alsever's solution (MP Biomedical, LLC) until the supernatant was clear, then counted and diluted to 1×10^8 cells per mL in (**HBSS++**) which consisted of HEPES buffered saline (150 m*M* NaCl, 4 m*M* $MgCl_2$, 10 m*M* HEPES, pH 7.2) with 0.01% gelatin (G9391, Sigma Aldrich) and 1 m*M* EGTA (AnaSpec). To pour agarose plates embedded with guinea pig RBC, 2% agarose solution was warmed to 56°C, and HBSS++ and RBC were warmed to 45°C. Warmed agarose (8.2 mL) was pipetted into 6.5 mL of HBSS++ and cooled to 45°C before 4.1 mL of RBC were added to the agarose and HBSS++ mixture. This solution was mixed by swirling and quickly poured onto a pre-warmed 150-mm culture dish (#10062–882, VWR). To aid in pouring and mixing, all serological pipettes and dishes were pre-warmed in an incubator at 37°C. The poured plate was allowed to cool for 30 min at room temperature before the plate was cooled for 30 min at 4°C. Sample wells were made using a 5-mm biopsy punch and removal of the agarose punch by aspiration. To measure complement activity, 20 μL of sample were carefully pipetted into the sample well. Bovine serum diluted 2-fold in PBS and diluted 32-fold

in PBS were included as high and low positive controls. Heat-inactivated fetal bovine serum (65°C for 30 min) and heat-inactivated casein-depleted colostrum whey (65°C for 30 min) were included as negative controls on every plate. Plates were wrapped in parafilm and incubated overnight at 4°C. The following day, parafilm was removed and plates were incubated at 37°C for 90 min. The diameter of the hemolysis ring for each sample was measured using a digital caliper. All samples were assayed in duplicate.

Statistical Analysis

Outcomes of the experiment were differences in colostral leukocyte proportions and viability, microRNA abundance, and complement activity between RAW, HT, and FR colostrum, and colostrum, transition milk, and mature milk. In the absence of prior data to guide sample size calculation, we selected a minimum sample size of 12 for each sample type in both objectives. At a chosen level of $\alpha = 0.05$ we determined that this would give us power ≥ 0.85 to detect a change in proportion of viable leukocytes or leukocyte proportions by 0.5 (GPower v. 3.1.9.7).

Descriptive data on the sample populations, colostrum Brix percentage, and colostrum volume for each experiment were summarized by PROC UNIVARIATE of SAS (v. 9.4, SAS Institute Inc.). Treatment differences for outcomes measured in RAW, HT, and FR were investigated using mixed effects ANOVA in PROC MIXED of SAS using a model that included fixed effect of treatment, random effect of colostrum batch, and LSMEANS was used to test pair-wise comparisons of RAW versus 0 for cell viability. Longitudinal data on a subset of animals (n = 13) with a complete set of colostrum, transition milk, and mature milk samples were analyzed by repeated measures ANOVA using PROC MIXED of SAS with fixed effect of time and REPEATED statement for time, subject of cow, and first-order autoregressive or heterogeneous first-order autoregressive covariance structure depending on the lowest Akaike information criterion. Familywise error rate of multiple pair-wise comparisons was adjusted using the Bonferroni correction. Normality and homoscedasticity of residuals were visually assessed for all models. To fulfill model assumptions the dependent variable total bacterial counts were log-transformed and reported as the geometric mean and back-transformed 95% confidence interval (CI). Somatic cell count linear score (**LS**) was calculated as [log2(SCC/100,000) + 3]. The INFLUENCE statement was used to test for outliers using Cook's distance defined as a value >0.5. Values are reported as least squares means and standard error of the mean unless otherwise noted.

RESULTS

Experiment 1

Colostrum was collected from a total of 14 cows, with 4, 3, 4, and 4 cows in parities 1, 2, 3, and ≥4, respectively. The median (range) Brix percentage in colostrum at harvest was 25.8 (20.7 to 30.7) %. Colostrum Brix percentage in RAW, HT, and FR were 25.4 ± 0.8, 25.5 ± 0.8, and 25.3 ± 0.8%, respectively ($P = 0.11$). Colostrum yield averaged (±SD) 5.2 ± 3.7 L. Results of total bacterial counts in RAW, HT, and FR are presented in Figure 1. Total bacterial count geometric mean (back-transformed 95% CI) in RAW was 5,045 (2,070 to 12,297) cfu/mL and did not differ from FR [5,300 (2,174 to 12,918) cfu/mL, $P = 0.77$], but were decreased to 0 cfu in HT in 13 of 14 samples. Cultures for *Streptococcus agalactiae*, *Klebsiella* spp., gram-positive *Bacillus* spp., *Corynebacterium* spp., *Trueperella pyogenes*, and other organisms (*Prototheca* spp., yeast, mold, fungi) were negative for all samples. *Streptococcus* spp., *Staphylococcus* spp., other coliforms, and gram-negative *Bacillus* spp. were detected in RAW and FR samples. *Escherichia coli* were detected in 3 FR samples, but not their corresponding RAW samples. *Pseudomonas* spp. were detected in 3 samples of RAW but not their corresponding FR, and detected in 2 samples of RAW but not the corresponding FR samples. *Staphylococcus aureus* was detected in 1 RAW sample and the corresponding FR sample. One HT sample showed 1,780 cfu/mL and was positive for *Streptococcus* spp. and *Staphylococcus* spp.

The proportion of viable cells isolated from RAW is presented in Figure 1. Viable cells were not isolated from HT and FR. Manual differential cell counts identified macrophages, neutrophils, and lymphocytes as the primary cell types in RAW. In RAW, macrophages or neutrophils made up the greatest proportions of leukocytes and lymphocytes the smallest. Macrophages, neutrophils, and lymphocytes averaged ± SD (range) 53 ± 30 (11 to 95), 30 ± 24 (3 to 71), and 17 ± 14 (1 to 46) %, respectively. Overall cell preservation in cytocentrifuge slides for all HT and FR, as well as one RAW sample was too poor to perform a reliable cell differential. Representative cytology images obtained from cytocentrifuge preparations of RAW, HT, and FR colostrum samples are presented in Figure 1. In cytology images with good cell preservation, macrophages and neutrophils were observed that were vacuolated in appearance with clear discrete, lipid-like vacuoles. Neutrophils also frequently contained smooth blue material within their cytoplasm, which was presumed to be milk protein (see Figure 3A for example). Anucleate keratinized squamous epithelial

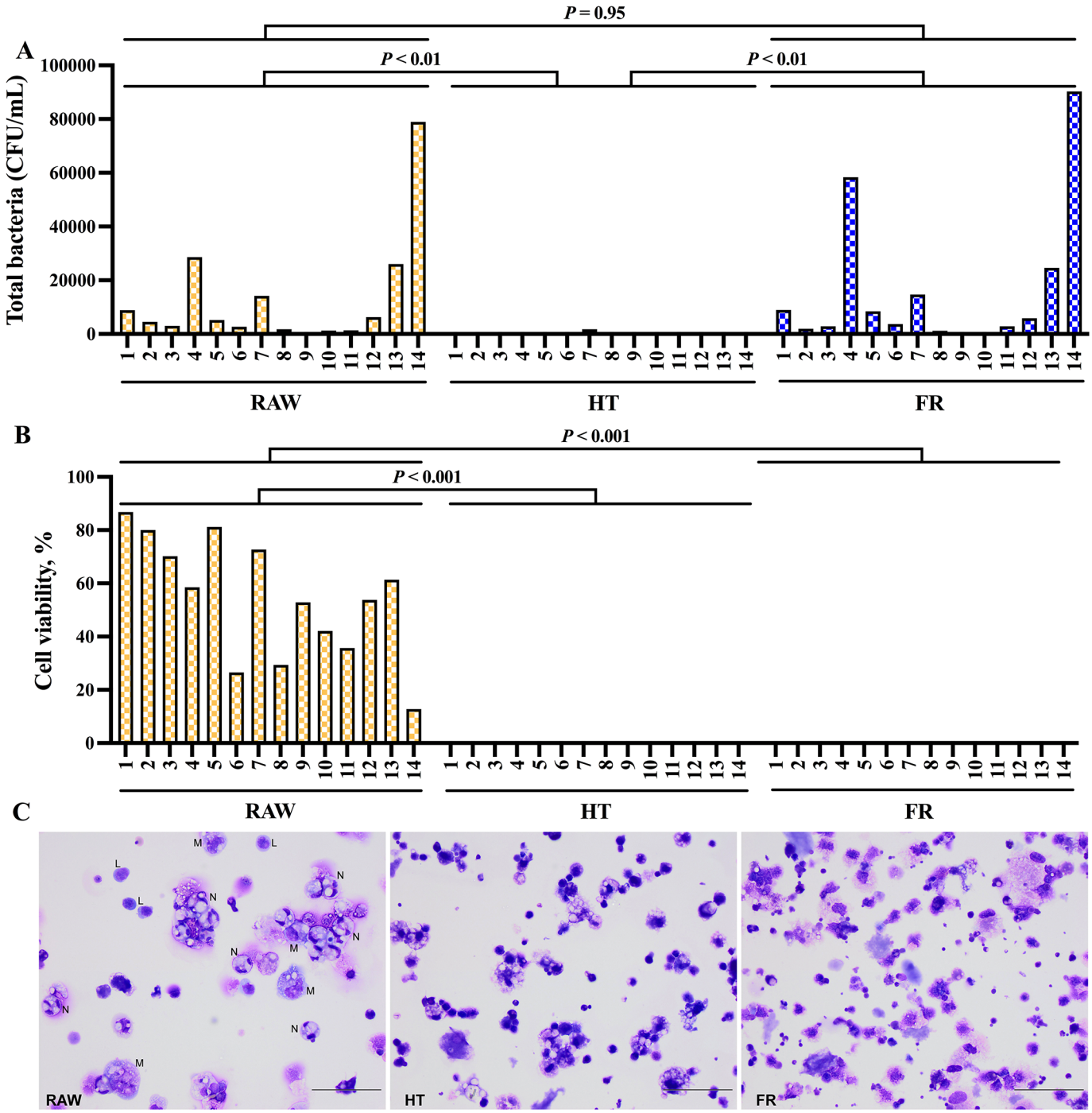


Figure 1. Total bacteria count (A), viability of isolated cells (B) from colostrum from 14 Holstein cows entering their first to sixth parity, and cytology images (C) from an individual animal as an example of a colostrum sample either cooled on ice following harvest (RAW, n = 14) and stored at 4°C overnight, heat-treated for 60 min at 60°C (HT, n = 14) before being cooled on ice and stored at 4°C overnight, or frozen at −20°C overnight (FR, n = 14). Data analyzed by ANOVA and shown with Tukey's adjustment for pair-wise comparison. Data presented for individual samples. Leukocytes include macrophages (M), neutrophils (N), and lymphocytes (L). Wright's stain. Scale bars = 50μm.

cells were occasionally present and interpreted as skin contaminants. Nucleated epithelial cells resembling mammary glandular epithelium were not seen. Occasional cytocentrifuge preparations from raw samples contained intracellular and extracellular bacterial cocci and rods, as well as rare yeast.

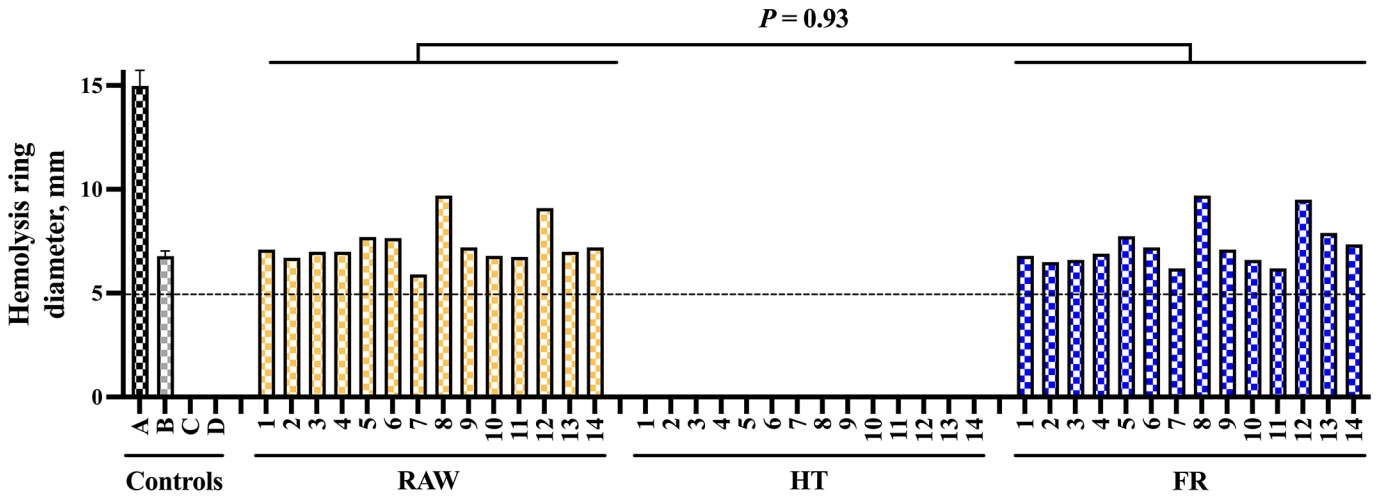


Figure 2. Hemolysis activity of the alternative complement pathway from colostrum from 14 Holstein cows entering their first or sixth parity that was either cooled on ice following harvest (RAW) and stored at 4°C overnight, heat-treated for 60 min at 60°C (HT) before being cooled on ice and stored at 4°C overnight, or frozen at −20°C overnight (FR), and mean and SD of positive and negative controls included on 4 assay plates. Alternative complement pathway activity was determined by semi-solid hemolysis assay and is presented as hemolysis ring diameter. Controls included (A) bovine serum diluted 2-fold in PBS, (B) bovine serum diluted 32-fold in PBS, (C) heat-inactivated (HI, 65°C for 30 min) fetal bovine serum (FBS), and (D) HI whey that was depleted of casein by ultracentrifugation. No hemolysis was observed for HI controls. Dashed horizontal line represents the lower limit of detection of the hemolysis assay.

The abundance of miRNA isolated from RAW, HT, and FR was 10.2 ± 3.0, 3.9 ± 3.0, and 18.6 ± 3.0 ng, respectively. Abundance of miRNA did not differ between RAW and HT ($P = 0.17$), but was increased ($P < 0.05$) in FR compared with RAW and HT and increased ($P < 0.001$) in FR compared with HT.

Alternative complement pathway activity did not differ ($P = 0.93$) between RAW and FR, but activity was below the lower limit of detection of the assay for all HT (Figure 2). Interassay CV of the complement pathway assay was 4.4%, and hemolysis activity of controls included on each assay plate is presented in Figure 2.

Experiment 2

Colostrum was collected from a total of 23 cows, with 9, 11, and 3 cows of parities 2, 3, and 4. Average (range) time to colostrum harvest after calving was 5 h (1 to 8 h). Longitudinal milk samples were collected from 13 cows, with 7 and 3 cows of parities 2 and 3. The median (range) Brix percentage from 23 colostrum samples was 26.8 (20.8 to 37.1), and the average (±SD) colostrum weight was 8.2 ± 3.5 kg. Average (range) SCC LS in colostrum samples (n = 23) was 6.9 (4.5 to 8.8). Somatic cell count LS in the subset of longitudinal samples of colostrum, transition milk, and mature milk is presented in Figure 3. Somatic cell count LS was greatest in colostrum and decreased in transition and mature milk. Average ± SD (range) viability of cells isolated from colostrum samples (n = 23) was 88 ± 4 (80 to 96) %. Viability of cells isolated from a subset of longitudinal samples of colostrum, transition milk, and mature milk did not differ over time ($P = 0.63$). Viability of isolated cells in colostrum, transition milk, and mature milk was 89.3 ± 1.5, 91.4 ± 1.5, and 89.3 ± 1.5%, respectively.

Manual cell differentials identified macrophages, neutrophils, and lymphocytes as the primary cell types in colostrum and milk samples. Macrophages were vacuolated in appearance, and often contained clear discrete lipid-like droplets. One sample contained multinucleated giant cells, as well as larger foamy macrophages. Eosinophils were rarely identified. Anucleate keratinized squamous epithelial cells were occasionally present. Several samples contained bacteria and one sample contained fungal hyphae and bacteria. In colostrum samples (n = 23), proportions of cell types were highly variable with macrophages most often making up the greatest proportion of leukocytes with an average ± SD (range) 51.9 ± 20.0 (14 to 89) %. Average ± SD (range) neutrophil proportion was 35.1 ± 17.1% (6 to 68%). Lymphocytes made up the smallest proportion of leukocytes with an average ± SD (range) of 13.0 ± 8.5% (2 to 37%). The proportion of cell types isolated from a subset of longitudinal samples of colostrum, transition milk, and mature milk is presented in Figure 3. Macrophages made up the greatest proportion of leukocytes in colostrum, transition milk, and mature milk, but did not differ as colostrum transitioned to mature milk ($P = 0.52$). The

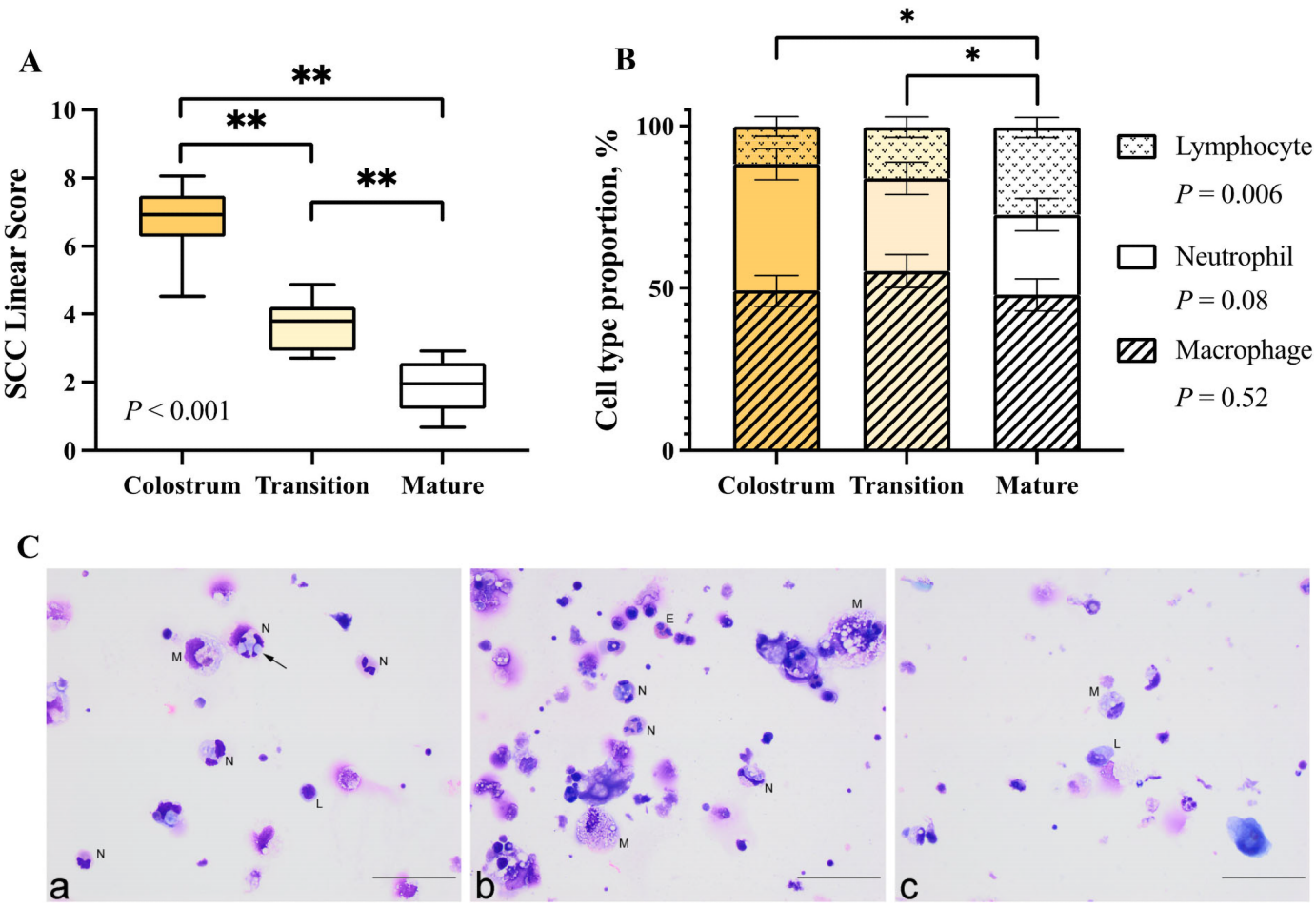


Figure 3. Box and whisker plot representing SCC linear score (A) and proportions of leukocyte cell types from manual differential cell counts of isolated cells (B) from colostrum (first milking ≤8 h after parturition), transition milk (3 to 4 d postpartum), and mature milk (6 to 7 d postpartum) collected from 13 multiparous Holstein cows entering their second or third parity. Cytology images (C) from (a) colostrum, (b) transition milk, and (c) mature milk from one animal. *P*-value for effect of time shown. Data analyzed by ANOVA and shown with Tukey's adjustment for pair-wise comparisons (*) $P < 0.05$, (**) $P < 0.001$. Box and whisker plots represent median (midline), interquartile range (upper and lower edges of boxes), and minimum and maximum (whiskers). Data presented as LSM ± SE for cell proportion. Leukocytes include macrophages (M), neutrophils (N), lymphocytes (L), and rare eosinophils (E). Arrow points to smooth blue material frequently found within neutrophils (presumptive milk protein). Wright's stain. Scale bars = 50 μm.

proportion of lymphocytes increased more than 2-fold ($P = 0.006$) in mature milk compared with transition milk and colostrum. Representative cytology images obtained from cytocentrifuge preparations of colostrum, transition milk, and mature milk collected from the same animal is presented in Figure 3.

MicroRNA was isolated from longitudinal samples collected from colostrum, transition milk, and mature milk of a subset of cows (n = 13). The abundance of miRNA isolated from longitudinal samples is presented in Figure 4. The abundance of miRNA isolated decreased ($P = 0.03$) from colostrum to transition and mature milk. The CT for microRNA of interest in colostrum, transition milk, and mature milk are presented in Figure 4. The CT did not differ from colostrum to transition and mature milk for miR-148a-3p ($P = 0.49$), miR-27b-3p ($P = 0.39$), miR-let-7a-5p ($P = 0.24$), and miR-26a-5p ($P = 0.29$). The number of samples at each time point that had CT ≥35 or had no amplification of target are presented in Table 1. Electropherogram patterns of the size distribution of miRNA pooled from colostrum, transition milk, and mature milk is presented in Figure 5.

Alternative complement pathway activity was observed in colostrum, but activity was below the lower limit of detection of the assay for transition and mature milk (Figure 6). Interassay CV of the complement pathway assay was 4.5%.

Table 1. Number of colostrum, transition milk, and mature milk samples with microRNA (miRNA) cycle threshold (CT) ≥35 or no amplification

Target	Sample[1]	CT ≥35	No amplification
miR-148a-3p	Colostrum	1	1
	Transition	3	
	Mature	2	2
miR-27b-3p	Colostrum		
	Transition		
	Mature	1	
miR-let-7a-5p	Colostrum		
	Transition		
	Mature		1
miR-26a-5p	Colostrum	1	
	Transition		5
	Mature	1	4

[1]Samples were collected from colostrum (first milking ≤8 h from parturition), transition (3 to 4 d postpartum), and mature milk (6 to 7 d postpartum) from 13 multiparous Holstein cows entering their second or third parity.

DISCUSSION

Our first objective was to study the effects of heat-treatment and freezing on the viability of colostral leukocytes, abundance of microRNA, and activity of the alternative complement pathway as compared with raw colostrum. As expected, heat treatment at 60°C for 60 min decreased total bacterial counts to 0 in nearly all colostrum samples as we have previously observed (Mann et al., 2020). All the raw samples were substantially below current industry goals for maximum counts of total bacteria, total coliform, streptococci, and staphylococci (McGuirk and Collins, 2004). Long-term frozen storage (≥1 wk) of raw bovine milk samples has been associated with species-dependent changes in bacterial counts and the number of samples culture positive for specific pathogens (Schukken et al., 1989; Alrabadi, 2015; Gille et al., 2018), but total bacterial counts are generally unchanged or decreased with long-term freezing. We did not identify studies that have evaluated overnight, or short-term cold storage on total bacterial counts in bovine colostrum. In the current study, overnight freezing was not associated with a change in total bacterial counts obtained from culture.

Using the centrifugation and washing protocol described here, we were able to isolate viable cells from diluted RAW colostrum with similar proportions of bovine colostral leukocytes described previously using electron microscopy or cytocentrifuge preparations and differential cell counts (Lee et al., 1980, Park et al., 1992, Ellis et al., 1996). These proportions were consistent with leukocyte proportions reported for human colostrum (Crago et al., 1979; Hassiotou and Geddes, 2015). We acknowledge that cell isolation by centrifugation biases cell retrieval toward intact and viable cells (Axelsson, 2010), but it is unknown if this would bias cell type isolation as well. We identified lymphocytes in colostrum, but at comparatively lower proportions than macrophages and neutrophils. As neutrophils and macrophages are recruited to the mammary gland upon detection of pathogens these cell types may be present to primarily benefit the dam (Paape et al., 1977). Given the paucity of data confirming immune competence of colostral lymphocytes and their mechanisms to influence neonatal immunity, it remains to be determined if these cells functionally benefit the neonate (Ellis, 2021). The high lipid content of colostrum, intracellular lipid droplets, and the phagocytosis of fat globules are thought to limit the efficiency of leukocyte isolation, particularly lymphocytes, due to altered buoyant den-

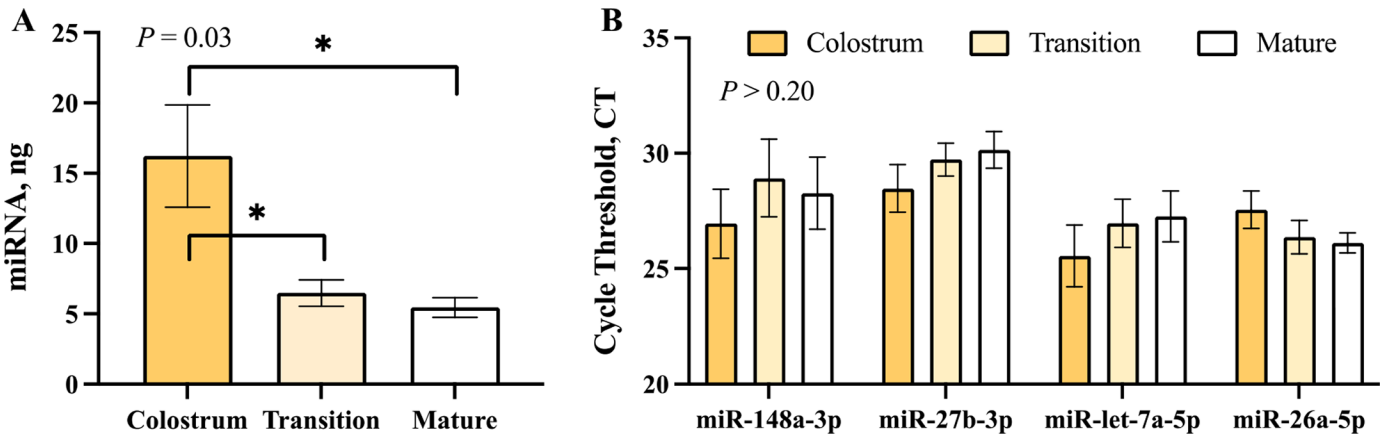


Figure 4. Total microRNA (miRNA) extracted from extracellular vesicles isolated by ultracentrifugation from 6 mL of colostrum and milk (A), and cycle threshold (CT) during real-time reverse-transcription quantitative PCR for selected miRNA targets extracted from extracellular vesicles (B) from colostrum (first milking ≤8 h after parturition), transition milk (3 to 4 d postpartum), and mature milk (6 to 7 d postpartum) collected from 13 multiparous Holstein cows entering their second or third parity. *P*-value for effect of time shown. Data analyzed by ANOVA and shown with Tukey's adjustment for pair-wise comparisons (*) $P < 0.05$. Data presented as LSM ± SE.

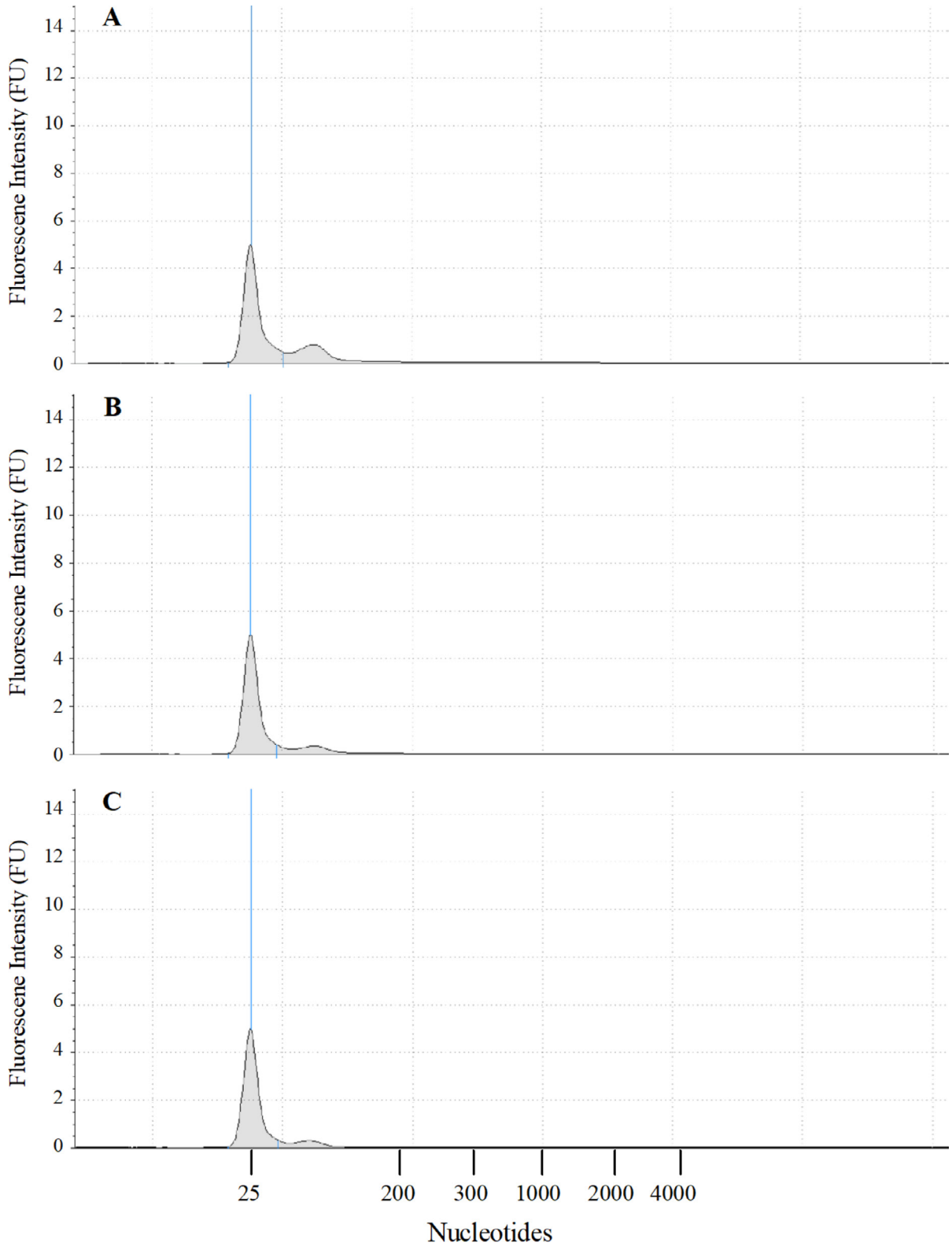


Figure 5. Electropherogram patterns of size distribution of RNA extracted from extracellular vesicles isolated by ultracentrifugation and composited from (A) colostrum (first milking ≤8 h after parturition), (B) transition milk (3 to 4 d postpartum), and (C) mature milk (6 to 7 d postpartum) collected from 13 multiparous Holstein cows entering their second or third parity. An equal volume of RNA was pooled from each sample after extraction for colostrum, transition milk, and mature milk before analysis using a Tape-Station system and RNA Screentape Assay (Agilent Technologies). Vertical blue lines represent small RNA peak in the lower molecular weight region.

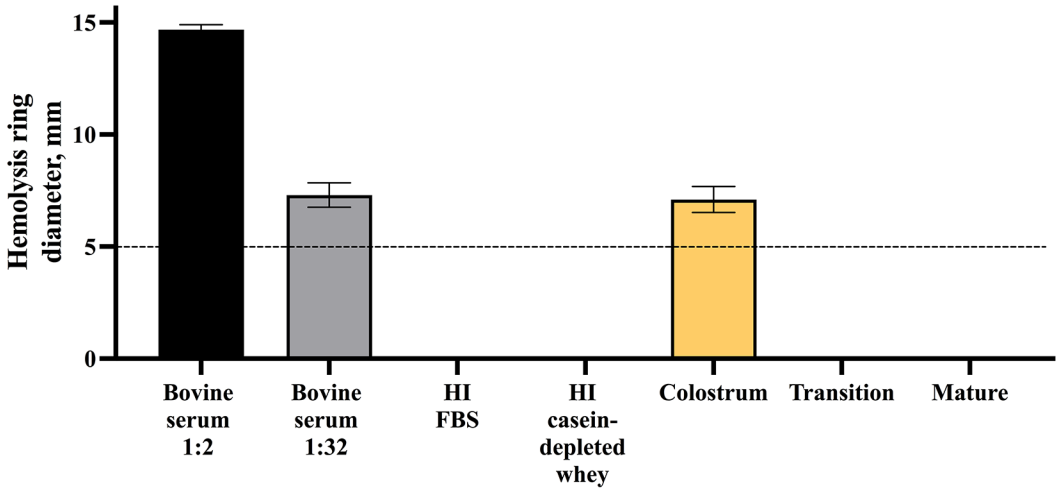


Figure 6. Least squares means and SE of hemolysis activity of the alternative complement pathway from colostrum (first milking ≤8 h after parturition), transition milk (3 to 4 d postpartum), and mature milk (6 to 7 d postpartum) collected from 13 multiparous Holstein cows entering their second or third parity, and the mean and SD of hemolysis activity from positive and negative controls included on 4 assay plates. Alternative complement pathway activity was determined by semi-solid hemolysis assay and is presented as hemolysis ring diameter. Controls included bovine serum diluted 2-fold in PBS, bovine serum diluted 32-fold in PBS, heat-inactivated (HI, 65°C for 30 min) fetal bovine serum (FBS), and HI whey that was depleted of casein by ultracentrifugation. No hemolysis was observed for HI controls. Dashed horizontal line represents the lower limit of detection of the hemolysis assay.

sity (Meganck et al., 2014). We did observe fat-laden or foamy macrophages, as have been described in human colostrum (Smith and Goldman, 1968; Crago et al., 1979); however, smears from the fat layer removed after centrifugation were stained and confirmed the absence of cells (data not shown), suggesting that cells were not lost in the buoyant fat layer. We did not observe contaminating mammary epithelial cells in cytocentrifuge preparations from colostrum as has been similarly reported (Lee et al., 1980).

Along with reducing bacterial contamination, the heat-treatment protocol eliminated viable cells in the present study. Godden et al. (2012) reported reduced viable cell counts following heat-treatment of colostrum, but SCC measured by NIR assay was not affected by heat-treatment. Somatic cell count measurements, including optical fluorescence, still detect most the colostral cellular components even if not viable after heat-treatment (Godden et al., 2012) or freezing (Barkema et al., 1997). We previously reported that colostrum heat-treatment was associated with a decline in optical fluorescence-measured SCC with a median (range) reduction of 36 (0 to 90) % (Mann et al., 2020). Somatic cell counts do not accurately reflect viable colostral leukocytes and were therefore not measured in the experiment evaluating effects of heat-treatment and freezing on colostrum. Freezing colostrum has been used in some experiments as a method to eliminate maternal leukocytes to compare calf outcomes after feeding colostrum with and without viable cells (Costa et al., 2017; Martin et al., 2021). We confirmed that freezing also eliminated viable cells, but intact cells and cell components were still visible in cell cytocentrifuge preparations indicating that freezing colostrum does not produce a cell-free product. While some data suggest that maternal leukocytes can be translocated across the gut epithelial barrier and influence immune development of the neonate (Reber et al., 2008), heat-treatment and freezing likely eliminate biological cellular activity in the newborn calf.

We isolated miRNA from colostrum using a similar ultracentrifugation protocol to sediment extracellular vesicles as previously reported (Hata et al., 2010; Sun et al., 2013; Izumi et al., 2015). Work by Izumi et al. (2012) demonstrated that extracellular vesicle-miRNA isolated from raw, mature bovine milk whey was resistant to heating, RNase treatment, and exposure to an acidic environment. Storage in extracellular vesicles potentially makes miRNA available to regulate gene expression within the neonate (Sun et al., 2013), and likely protected miRNA during heat-treatment here. The notable numerical decrease in miRNA abundance of HT compared with RAW may be an artifact of a change in fluidity characteristics of heat-treated colostrum as previously described (McMartin et al., 2006) rather than a direct effect on miRNA abundance. Previous work isolated whey before heat-treatment (Izumi et al., 2012). Freezing whole colostrum in the current experiment may have led to an increase in miRNA abundance following the lysing of cells and their re-

lease of vesicles containing miRNA. We do not know if the gastrointestinal abundance and activity of miRNA would be similarly altered in calves following the ingestion of heat-treated or frozen colostrum. A limitation of the method used here was that we did not characterize the proteome of isolated vesicles to confirm the presence of exosomes.

We have previously shown that heat-treatment of high quality colostrum did not affect IgG concentration, but decreased concentration of IgA and the abundance of several low-abundant proteins in the complement pathway detected by proteomics (Mann et al., 2020). The current study extends these findings by investigating the alternative complement pathway activity in colostrum which we confirmed was decreased after heat treatment. Complement factors have the potential to play beneficial biological roles within the gut lumen by eliminating pathogens through microorganism killing and opsonization (Korhonen et al., 2000, Ricklin et al., 2010), for example, bactericidal complement activity of colostrum has been demonstrated but was decreased following heat-treatment (Brock et al., 1975). However, the biological significance of the observed change to the neonate requires further investigation. We were specifically interested in alternative complement pathway activity in the current study because we reported that heat-treatment had prominent effects on alternative pathway complement components (Mann et al., 2020). The alternative pathway is activated directly by antigen in the absence of antibodies (Ogundele, 1999) which would be of specific benefit to the naïve bovine neonate, as opposed to the classical pathway that is activated mainly by Ig complexes. To our knowledge, this is the first study using a semi-solid hemolysis assay to measure alternative complement pathway activity in bovine colostrum. In the samples here, alternative complement pathway activity was variable but similar to activity of bovine serum after 32-fold dilution. Differences in methodology between this and previous experiments make a direct comparison difficult, but Eckblad et al. (1981) used a 50% hemolytic complement activity (CH50) assay and reported that colostral whey total complement activity was 2.2% of total complement activity in serum collected from the same animal emphasizing the comparatively lower complement activity in colostrum.

The bovine neonate likely gains immune support through a multitude of diverse immunologically active components that are transferred through colostrum. As these factors continue to be characterized and their biological role interpreted, more work will be needed to determine their viability in colostrum undergoing different postharvest treatments and storage conditions. Data from this experiment suggest that heat-treatment and freezing alter the abundance and activity of some factors that may be beneficial to the neonate.

Our second objective was to characterize maternal leukocytes, miRNA, and complement activity as colostrum transitions to mature milk. A novel aspect of our study was characterizing leukocyte proportions as colostrum transitions to mature milk. We noted similar proportions of macrophages, neutrophils and lymphocytes in colostrum from both experiments. Despite decreased numbers of maternal cells in transition and mature milk, indicated by decreased SCC LS, viability of isolated colostral cells in experiment 2 was maintained in transition and mature milk and similar to viability of cells isolated from colostrum reported by others (Nonnecke and Kehrli, 1985; Meganck et al., 2014). Animals sampled in this experiment did not present with signs of clinical mastitis at the time of colostrum harvest and exponential decreases in SCC LS in transition and mature milk are indicative of the absence of obvious inflammation in the mammary gland.

Similar to colostral leukocytes and other colostral factors including Ig, microRNA and complement activity were enriched in colostrum compared with transition and mature milk. In a previous study, total complement activity, which would include alternative pathway activity, was undetected in milk taken 96 h or one month after parturition (Eckblad et al., 1981). Advancements in characterizing cell-free nucleic acids have identified specific microRNA enriched in colostrum (Sun et al., 2013; Ylioja et al., 2019; Van Hese et al., 2020), and in the current study we compared the abundance of miRNA targets that have been identified as immune-related and conserved in several mammalian species including cows, sheep, and humans (Yun et al., 2021). The profile of miRNA targets we investigated through PCR appeared to be maintained in total miRNA despite overall reduced miRNA isolated from an equal volume of colostrum, transition milk, and mature milk. Though not examined here, heat-treatment of transition milk would also likely eliminate viable cells and alter complement activity, limiting the biological role of these components in calves fed heat-treated transition milk.

CONCLUSIONS

We conclude that non-Ig immunologically active factors in colostrum, including maternal leukocytes, microRNA, and complement pathway activity, are sensitive to heat treatment and freezing. We further conclude that these factors are variable in abundance as colostrum transitions to mature milk. Future studies will have to clarify the benefit of non-Ig factors to the immunologically-naïve developing neonate that are

altered postharvest by heat-treatment or freezing, and continue to be present in detectable quantities in transition and mature milk.

ACKNOWLEDGMENTS

We gratefully acknowledge the commercial dairy farm and farm personnel who allowed us to carry out this project, as well as the staff at the Cornell University Ruminant Center (Harford, NY). We thank Suzanne Klaessig (Cornell University, Ithaca, NY) for assistance with sample preparation and Julia Felippe (Cornell University, Ithaca, NY) for use of the cytocentrifuge equipment. The authors also thank Brian Rudd (Cornell University) for technical expertise and Praveen Sethupathy and Mike Shanahan (Cornell University) for use of the Tape-Station system. This work was supported by the USDA National Institute of Food and Agriculture Hatch/Multistate project no. 2020-21-143 (Washington, DC). Any opinions, findings, conclusions, or recommendations expressed in this publication are those of the author(s) and do not necessarily reflect the view of the National Institute of Food and Agriculture of the USDA. The authors have not stated any conflicts of interest.

REFERENCES

Alrabadi, N. 2015. The effect of freezing on different bacterial counts in raw milk. Int. J. Biol. 7:9–12. https://doi.org/10.5539/ijb.v7n4p9.

AOAC International. 2012. Official Methods of Analysis. 19th ed. AOAC International.

Axelsson, H. 2010. Cell separation, centrifugation. Pages 1427–1445 in Encyclopedia of Industrial Biotechnology. M. C. Flickinger, ed. https://doi.org/10.1002/9780470054581.eib203.

Barkema, H. W., J. Van Der Schans, Y. H. Schukken, A. L. W. De Gee, T. J. G. M. Lam, and G. Benedictus. 1997. Effect of freezing on somatic cell count of quarter milk samples as determined by a Fossomatic electronic cell counter. J. Dairy Sci. 80:422–426. https://doi.org/10.3168/jds.S0022-0302(97)75953-8.

Brock, J. H., F. Ortega, and A. Pineiro. 1975. Bactericidal and haemolytic activity of complement in bovine colostrum and serum: Effect of proteolytic enzymes and ethylene glycol tetraacetic acid (EGTA). Ann. Immunol. Paris 126C:439–451.

Cabinian, A., D. Sinsimer, M. Tang, O. Zumba, H. Mehta, A. Toma, D. Sant'Angelo, Y. Laouar, and A. Laouar. 2016. Transfer of maternal immune cells by breastfeeding: Maternal cytotoxic T lymphocytes present in breast milk localize in the Peyer's patches of the nursed infant. PLoS One 11:e0156762. https://doi.org/10.1371/journal.pone.0156762.

Costa, J. F. D. R., S. M. F. Novo, C. C. Baccili, N. M. Sobreira, D. J. Hurley, and V. Gomes. 2017. Innate immune response in neonate Holstein heifer calves fed fresh or frozen colostrum. Res. Vet. Sci. 115:54–60. https://doi.org/10.1016/j.rvsc.2017.01.008.

Crago, S. S., S. J. Prince, T. G. Pretlow, J. R. McGhee, and J. Mestecky. 1979. Human colostral cells I. Separation and characterization. Clin. Exp. Immunol. 38:585–597.

Donahue, M., S. M. Godden, R. Bey, S. Wells, J. M. Oakes, S. Sreevatsan, J. Stabel, and J. Fetrow. 2012. Heat treatment of colostrum on commercial dairy farms decreases colostrum microbial counts while maintaining colostrum immunoglobulin G concentrations. J. Dairy Sci. 95:2697–2702. https://doi.org/10.3168/jds.2011-5220.

Donovan, D. C., A. J. Reber, J. D. Gabbard, M. Aceves-Avila, K. L. Galland, K. A. Holbert, L. O. Ely, and D. J. Hurley. 2007. Effect of maternal cells transferred with colostrum on cellular responses to pathogen antigens in neonatal calves. Am. J. Vet. Res. 68:778–782. https://doi.org/10.2460/ajvr.68.7.778.

Eckblad, W. P., K. M. Hendrix, and D. P. Olson. 1981. Total complement hemolytic activity of colostral whey and sera from dairy cows. Cornell Vet. 71:54–58.

Ellis, J. 2021. Passive transfer of colostral leukocytes: A benefit/risk analysis. Can. Vet. J. 62:233–239.

Ellis, J. A., L. E. Hassard, V. S. Cortese, and P. S. Morley. 1996. Effects of perinatal vaccination on humoral and cellular immune responses in cows and young calves. J. Am. Vet. Med. Assoc. 208:393–400.

Ghosh, M. K., V. Nguyen, H. K. Muller, and A. M. Walker. 2016. Maternal milk T cells drive development of transgenerational Th1 immunity in offspring thymus. J. Immunol. 197:2290–2296. https://doi.org/10.4049/jimmunol.1502483.

Gille, L., F. Boyen, L. Van Driessche, B. Valgaeren, F. Haesebrouck, P. Deprez, and B. Pardon. 2018. Short communication: Effect of freezer storage time and thawing method on the recovery of *Mycoplasma bovis* from bovine colostrum. J. Dairy Sci. 101:609–613. https://doi.org/10.3168/jds.2017-13130.

Godden, S. M., K. Hausler, N. Koewler, K. Brakefield, H. Lillegaard, M. Van Amburgh, R. Bey, and P. Rapnicki. 2012. Effect of heat-treatment on nutritional and immune factors in bovine colostrum. Minnesota Dairy Health Conference, May 22nd to 24th, Minneapolis, MN.

Godden, S. M., J. E. Lombard, and A. R. Woolums. 2019. Colostrum management for dairy calves. Vet. Clin. North Am. Food Anim. Pract. 35:535–556. https://doi.org/10.1016/j.cvfa.2019.07.005.

Hassiotou, F., and D. T. Geddes. 2015. Immune cell-mediated protection of the mammary gland and the infant during breastfeeding. Adv. Nutr. 6:267–275. https://doi.org/10.3945/an.114.007377.

Hata, T., K. Murakami, H. Nakatani, Y. Yamamoto, T. Matsuda, and N. Aoki. 2010. Isolation of bovine milk-derived microvesicles carrying mRNAs and microRNAs. Biochem. Biophys. Res. Commun. 396:528–533. https://doi.org/10.1016/j.bbrc.2010.04.135.

Izumi, H., N. Kosaka, T. Shimizu, K. Sekine, T. Ochiya, and M. Takase. 2012. Bovine milk contains microRNA and messenger RNA that are stable under degradative conditions. J. Dairy Sci. 95:4831–4841. https://doi.org/10.3168/jds.2012-5489.

Izumi, H., M. Tsuda, Y. Sato, N. Kosaka, T. Ochiya, H. Iwamoto, K. Namba, and Y. Takeda. 2015. Bovine milk exosomes contain microRNA and mRNA and are taken up by human macrophages. J. Dairy Sci. 98:2920–2933. https://doi.org/10.3168/jds.2014-9076.

Kerboua, K. E., and K. Djenouhat. 2020. Semi-solid phase assay for the alternative complement pathway activity assessment (AP100). J. Immunoassay Immunochem. 41:144–151. https://doi.org/10.1080/15321819.2019.1696819.

Korhonen, H., P. Marnila, and H. S. Gill. 2000. Milk immunoglobulins and complement factors. Br. J. Nutr. 84(Suppl. 1):75–80. https://doi.org/10.1017/S0007114500002282.

Lee, C. S., F. B. P. Wooding, and P. Kemp. 1980. Identification, properties, and differential counts of cell populations using electron microscopy of dry cows secretions, colostrum, and milk from normal cows. J. Dairy Res. 47:39–50. https://doi.org/10.1017/S0022029900020860.

Mann, S., G. Curone, T. L. Chandler, P. Moroni, J. Cha, R. Bhawal, and S. Zhang. 2020. Heat treatment of bovine colostrum: I. Effects on bacterial and somatic cell counts, immunoglobulin, insulin, and IGF-I concentrations, as well as the colostrum proteome. J. Dairy Sci. 103:9368–9383. https://doi.org/10.3168/jds.2020-18618.

Martin, C. C., S. M. F. N. de Oliveira, J. F. R. Costa, C. C. Baccili, B. T. Silva, D. J. Hurley, and V. Gomes. 2021. Influence of feeding fresh colostrum from the dam or frozen colostrum from a pool on indicator gut microbes and the inflammatory response in neonatal calves. Res. Vet. Sci. 135:355–365. https://doi.org/10.1016/j.rvsc.2020.10.017.

McGuirk, S. M., and M. Collins. 2004. Managing the production, storage, and delivery of colostrum. Vet. Clin. North Am. Food Anim. Pract. 20:593–603. https://doi.org/10.1016/j.cvfa.2004.06.005.

McMartin, S., S. Godden, L. Metzger, J. Feirtag, R. Bey, J. Stabel, S. Goyal, J. Fetrow, S. Wells, and H. Chester-Jones. 2006. Heat treatment of bovine colostrum. I: Effects of temperature and viscosity and immunoglobulin G level. J. Dairy Sci. 89:2110–2118. https://doi.org/10.3168/jds.S0022-0302(06)72281-0.

Meganck, V., B. M. Goddeeris, E. Stuyven, S. Piepers, E. Cox, and G. Opsomer. 2014. Development of a method for isolating bovine colostrum mononuclear leukocytes for phenotyping and functional studies. Vet. J. 200:294–298. https://doi.org/10.1016/j.tvjl.2014.02.021.

Nonnecke, B. J., and M. E. Kehrli. 1985. Isolation of mononuclear cells from bovine milk by continuous-flow and density gradient centrifugation: Response to mitogens. Am. J. Vet. Res. 46:1259–1262.

Ogundele, M. O. 1999. Inhibitors of complement activity in human breast-milk: A proposed hypothesis of their physiological significance. Mediators Inflamm. 8:69–75. https://doi.org/10.1080/09629359990559.

Paape, M. J., R. E. Pearson, W. P. Wergin, and A. J. Guidry. 1977. Enhancement of chemotactic response of polymorphonuclear leukocytes into the mammary gland and isolation from milk. J. Dairy Sci. 60:53–62. https://doi.org/10.3168/jds.S0022-0302(77)83828-9.

Park, Y. H., L. K. Fox, M. J. Hamilton, and W. C. Davis. 1992. Bovine mononuclear leukocyte subpopulations in peripheral blood and mammary gland secretions during lactation. J. Dairy Sci. 75:998–1006. https://doi.org/10.3168/jds.S0022-0302(92)77842-4.

Reber, A. J., D. C. Donovan, J. Gabbard, K. Galland, M. Aceves-Avila, K. A. Holbert, L. Marshall, and D. J. Hurley. 2008. Transfer of maternal colostral leukocytes promotes development of the neonatal immune system Part II. Effects on neonatal lymphocytes. Vet. Immunol. Immunopathol. 123:305–313. https://doi.org/10.1016/j.vetimm.2008.02.009.

Reschke, C., E. Schelling, A. Michel, F. Remy-Wohlfender, and M. Meylan. 2017. Factors associated with colostrum quality and effects on serum gamma globulin concentrations of calves in Swiss dairy herds. J. Vet. Intern. Med. 31:1563–1571. https://doi.org/10.1111/jvim.14806.

Ricklin, D., G. Hajishengallis, K. Yang, and J. D. Lambris. 2010. Complement: A key system for immune surveillance and homeostasis. Nat. Immunol. 11:785–797. https://doi.org/10.1038/ni.1923.

Schukken, Y. H., J. A. H. Smit, F. J. Grommers, D. Vandegeer, and A. Brand. 1989. Effect of freezing on bacteriological culturing of mastitis milk samples. J. Dairy Sci. 72:1900–1906. https://doi.org/10.3168/jds.S0022-0302(89)79309-7.

Smith, C. W., and A. S. Goldman. 1968. The Cells of Human Colostrum. I. In Vitro Studies of Morphology and Functions. Pediatr. Res. 2:103–109. https://doi.org/10.1203/00006450-196803000-00005.

Sun, Q., X. Chen, J. Yu, K. Zen, C.-Y. Zhang, and L. Li. 2013. Immune modulatory function of abundant immune-related microRNAs in microvesicles from bovine colostrum. Protein Cell 4:197–210. https://doi.org/10.1007/s13238-013-2119-9.

Van Hese, I., K. Goossens, L. Vandaele, and G. Opsomer. 2020. Invited review: MicroRNAs in bovine colostrum-Focus on their origin and potential health benefits for the calf. J. Dairy Sci. 103:1–15. https://doi.org/10.3168/jds.2019-16959.

Ylioja, C. M., M. M. Rolf, L. K. Mamedova, and B. J. Bradford. 2019. Associations between body condition score at parturition and microRNA profile in colostrum of dairy cows as evaluated by paired mapping programs. J. Dairy Sci. 102:11609–11621. https://doi.org/10.3168/jds.2019-16675.

Yun, B., Y. Kim, D. J. Park, and S. Oh. 2021. Comparative analysis of dietary exosome-derived microRNAs from human, bovine and caprine colostrum and mature milk. J. Anim. Sci. Technol. 63:593–602. https://doi.org/10.5187/jast.2021.e39.

ORCIDS

T. L. Chandler https://orcid.org/0000-0001-6299-9436
A. Newman https://orcid.org/0000-0003-1990-8089
A. S. Sipka https://orcid.org/0000-0002-8646-824X
S. Mann https://orcid.org/0000-0003-1806-1154